

Q1 2021 (01 Sep Shift 2)

Let $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$. Let a vector $\vec{v}$ be in the plane containing $\vec{a}$ and $\vec{b}$. If $\vec{v}$ is perpendicular to the vector $3\hat{i} + 2\hat{j} - \hat{k}$ and its projection on $\vec{a}$ is 19 units, then $|2\vec{v}|^2$ is equal to .

Q2 2021 (31 Aug Shift 2)

Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors mutually perpendicular to each other and have same magnitude. If a vector $\vec{r}$ satisfies.

$$\vec{a} \times \{(\vec{r} - \vec{b}) \times \vec{a}\} + \vec{b} \times \{(\vec{r} - \vec{c}) \times \vec{b}\} + \vec{c} \times \{(\vec{r} - \vec{a}) \times \vec{c}\} = \vec{0},$$

then $\vec{r}$ is equal to :

(1) $\frac{1}{3}(\vec{a} + \vec{b} + \vec{c})$

(2) $\frac{1}{3}(2\vec{a} + \vec{b} - \vec{c})$

(3) $\frac{1}{2}(\vec{a} + \vec{b} + \vec{c})$

(4) $\frac{1}{2}(\vec{a} + \vec{b} + 2\vec{c})$

Q3 2021 (31 Aug Shift 1)

Let $\vec{a}$ and $\vec{b}$ be two vectors such that

$$|2\vec{a} + 3\vec{b}| = |3\vec{a} + \vec{b}| \text{ and the angle between } \vec{a} \text{ and } \vec{b} \text{ is } 60^\circ.$$

If $\frac{1}{8}\vec{a}$ is a unit vector, then $|\vec{b}|$ is equal to :

(1) 4

(2) 6

(3) 5

(4) 8

Q4 2021 (27 Aug Shift 1)

Let $\vec{a} = \hat{i} + 5\hat{j} + \alpha\hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} + \beta\hat{k}$ and $\vec{c} = -\hat{i} + 2\hat{j} - 3\hat{k}$ be three vectors such that, $|\vec{b} \times \vec{c}| = 5\sqrt{3}$ and $\vec{a}$ is perpendicular to $\vec{b}$. Then the greatest amongst the values of $|\vec{a}|^2$ is .

Q5 2021 (26 Aug Shift 2)

If the projection of the vector $\hat{i} + 2\hat{j} + \hat{k}$ on the sum of the two vectors $2\hat{i} + 4\hat{j} - 5\hat{k}$ and $-\lambda\hat{i} + 2\hat{j} + 3\hat{k}$ is 1, then λ is equal to .

Q6 2021 (26 Aug Shift 1)

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{j} - \hat{k}$. If $\vec{c}$ is a vector such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$, then $\vec{a} \cdot (\vec{b} \times \vec{c})$ is equal to :

(1) -2

(2) -6

(3) 6

(4) 2

Answer Key

Q1 (1494)

Q2 (3)

Q3 (3)

Q4 (90)

Q5 (5)

Q6 (1)

Q1 (1494)

$$\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$$

$$\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$$

$$\vec{c} = 3\hat{i} + 2\hat{j} - \hat{k}$$

$$\vec{v} = x\vec{a} + y\vec{b}$$

$$\vec{v} \cdot (3\hat{i} + 2\hat{j} - \hat{k}) = 0$$

$$\vec{v} \cdot \vec{a} = 19$$

$$\vec{v} = \lambda \vec{c} \times (\vec{a} \times \vec{b})$$

$$\vec{v} = \lambda [(\vec{c} \cdot \vec{b})\vec{a} - (\vec{c} \cdot \vec{a})\vec{b}]$$

$$= \lambda \left[(3 + 4 + 1)(2\hat{i} - \hat{j} + 2\hat{k}) - \left(\frac{6-2-2}{2} \right) (\hat{i} + 2\hat{j} + \hat{k}) \right]$$

$$= \lambda [16\hat{i} - 8\hat{j} + 16\hat{k} - 2\hat{i} - 4\hat{j} + 2\hat{k}]$$

$$\vec{v} = \lambda [14\hat{i} - 12\hat{j} + 18\hat{k}]$$

$$\lambda [14\hat{i} - 12\hat{j} + 18\hat{k}] \cdot \frac{(2\hat{i} - \hat{j} + 2\hat{k})}{\sqrt{4+1+4}} = 19$$

$$\lambda \frac{[28+12+36]}{3} = 19$$

$$\lambda \left(\frac{76}{3} \right) = 19$$

$$4\lambda = 3 \Rightarrow \lambda = \frac{3}{4}$$

$$|2\vec{v}| = \left| 2 \times \frac{3}{4} (14\hat{i} - 12\hat{j} + 18\hat{k}) \right|^2$$

$$= \frac{9}{4} \times 4(7\hat{i} - 6\hat{j} + 9\hat{k})^2$$

$$= 9(49 + 36 + 81)$$

$$= 9(166)$$

$$= 1494$$

Q2 (3)

$$\text{Suppose } \vec{r} = x\vec{a} + y\vec{b} + z\vec{c} \text{ and } |\vec{a}| = |\vec{b}| = |\vec{c}| = k$$

$$\vec{a} \times \{(\vec{r} - \vec{b}) \times \vec{a}\} + \vec{b} \times \{(\vec{r} - \vec{c}) \times \vec{b}\} + \vec{c} \times \{(\vec{r} - \vec{a}) \times \vec{c}\} = \vec{0}$$

$$\Rightarrow k^2(\vec{r} - \vec{b}) - k^2x\vec{a} + k^2(\vec{r} - \vec{c}) - k^2y\vec{b} + k^2(\vec{r} - \vec{a}) - k^2z\vec{c} = \vec{0}$$

$$\Rightarrow 3\vec{r} - (\vec{a} + \vec{b} + \vec{c}) - \vec{r} = \vec{0}$$

$$\Rightarrow \vec{r} = \frac{\vec{a} + \vec{b} + \vec{c}}{2}$$

Q3 (3)

$$|3\vec{a} + \vec{b}|^2 = |2\vec{a} + 3\vec{b}|^2$$

$$(3\vec{a} + \vec{b}) \cdot (3\vec{a} + \vec{b}) = (2\vec{a} + 3\vec{b}) \cdot (2\vec{a} + 3\vec{b})$$

$$9\vec{a} \cdot \vec{a} + 6\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{b} = 4\vec{a} \cdot \vec{a} + 12\vec{a} \cdot \vec{b} + 9\vec{b} \cdot \vec{b}$$

$$5|\vec{a}|^2 - 6\vec{a} \cdot \vec{b} = 8|\vec{b}|^2$$

$$5(8)^2 - 6 \cdot 8 \cdot |\vec{b}| \cos 60^\circ = 8|\vec{b}|^2 \left(\because \frac{1}{8}|\vec{a}| = 1 \right)$$

$$\Rightarrow |\vec{a}| = 8$$

$$40 - 3|\vec{b}| = |\vec{b}|^2$$

$$\Rightarrow |\vec{b}|^2 + 3|\vec{b}| - 40 = 0$$

$$|\vec{b}| = -8, |\vec{b}| = 5$$

(rejected)

Q4 (90)

$$\text{since, } \vec{a} \cdot \vec{b} = 0$$

$$1 + 15 + \alpha\beta = 0 \Rightarrow \alpha\beta = -16 \dots (1)$$

Also,

$$|\vec{b} \times \vec{c}|^2 = 75 \Rightarrow (10 + \beta^2)14 - (5 - 3\beta)^2 = 75$$

$$\Rightarrow 5\beta^2 + 30\beta + 40 = 0$$

$$\Rightarrow \beta = -4, -2$$

$$\Rightarrow \alpha = 4, 8$$

$$\Rightarrow |\vec{a}|_{\max}^2 = (26 + \alpha^2)_{\max} = 90$$

Q5 (5)

$$\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$$

$$\vec{b} = (2 - \lambda)\hat{i} + 6\hat{j} - 2\hat{k}$$

$$\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = 1, \vec{a} \cdot \vec{b} = 12 - \lambda$$

$$(\vec{a} \cdot \vec{b}) = |\vec{b}|^2$$

$$\lambda^2 - 24\lambda + 144 = \lambda^2 - 4\lambda + 4 + 40$$

$$20\lambda = 100 \Rightarrow \lambda = 5.$$

Q6 (1)

$$|\vec{a}| = \sqrt{3}; \vec{a} \cdot \vec{c} = 3; \vec{a} \times \vec{b} = -2\hat{i} + \hat{j} + \hat{k}, \vec{a} \times \vec{c} = \vec{b}$$

Cross with $\vec{a}$.

$$\vec{a} \times (\vec{a} \times \vec{c}) = \vec{a} \times \vec{b}$$

$$\Rightarrow (\vec{a} \cdot \vec{c})\vec{a} - a^2\vec{c} = \vec{a} \times \vec{b}$$

$$\Rightarrow 3\vec{a} - 3\vec{c} = -2\hat{i} + \hat{j} + \hat{k}$$

$$\Rightarrow 3\hat{i} + 3\hat{j} + 3\hat{k} - 3\vec{c} = -2\hat{i} + \hat{j} + \hat{k}$$

$$\Rightarrow \vec{c} = \frac{5\hat{i}}{3} + \frac{2\hat{j}}{3} + \frac{2\hat{k}}{3}$$

$$\therefore \vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c} = \frac{-10}{3} + \frac{2}{3} + \frac{2}{3} = -2$$